

(FRAM)-based 16-bit microcontroller series from TI. It includes a wide range of analogue and digital peripherals that makes configuration simple and reduces PCB size. FRAM combines the abilities of both RAM and ROM to ensure high speed, high endurance,

ultra-low power consumption, stability, flexibility, and more.

Reference designs provided by the company make it easier to retrofit the pre-existing water meters with automated meter reading (AMR) capabilities. With its ultrasonic sensing

feature, even small water leaks can be detected so that they get repaired rapidly. This not only benefits the dealer and consumer through intelligence and connectivity but also the environment by conserving water and energy.

—Ayushee Sharma

2 A ROBOT FOR PEOPLE WITH SPINAL DISABILITIES

Industry 4.0 technologies like robotics and machine learning find numerous uses in the medical field that directly or indirectly improve the lives of affected people. Capitalising on this potential, Bulgarian startup IRise Mechanics 357 is developing innovative robotic mobility devices for people with lower limb disabilities.

Lubomir Vassilevi, the founder of this company, decided to work on projects for people in need when his wife lost her ability to walk in an accident that resulted in paraplegia a few years ago. With the help of his team, he created the robotic tool Omnibot to not only improve the quality of life for his wife but also others suffering from such an issue.

Although aids like wheelchairs are available for people with motor disabilities, most of them require the persons to move in a sitting position. As the vertebral injuries take long to heal, and sometimes stay permanently for lifetime, using wheelchairs can lead to chronic secondary health issues like long-term pain, heart disease, osteoporosis, urinary tract infection, deep vein

thrombosis, and respiratory ailment.

Weighing just sixty kilograms, the chassis of Omnibot is built of aluminium tubing. It is lightweight, compact, and has foldable parts unlike other vertical devices available in the market. This makes the robot easy to transport in cars and other vehicles. It also enables easy transfer to and from wheelchair or any furniture and the machine without assistance.

One can easily grab objects on shelves and other hard-to-reach places with hands while on the bot. Its meca-num drive system (omnidirectional) enables 4 × 4 movement, axial movement and zero-grade turning radius. It works best for narrow indoor spaces but is unsuitable for overcoming obstacles or climbing stairs.

The device provides independence, decreases health risks and makes the person feel more socially included.

In testing phase, the bot was used by over thirty people having spinal cord injuries in Bulgaria and Austria. The device is planned to be sold at a retail price of € 21,000. It can be pre-ordered by paying € 1,000 to get thirty



per cent discount on price, according to the company. The startup was recently honoured with High Innovation Achievement award from President of the Republic of Bulgaria.

Besides being involved in creation of individualised aids, the team has also developed a motorcycle for people with lower limb disabilities and a wheelbarrow that is capable of running even marathons. The company is now looking for more to expand on a large scale, including foreign markets like Germany and the US.

—Ayushee Sharma

3 PREVENTING MACHINE FAILURE USING PREDICTIVE MAINTENANCE

When a major machinery breaks down because of electrical issues or component failure, you lose money every hour! This requires scalable, sustainable, and cost-effective monitoring to maximise runtime productivity and reduce expense on maintenance. Industry 4.0 is revolutionising manufacture and maintenance with the help of sensor-based technology that can

help save time, prevent costly downtime, and increase profits.

MachineSense, headquartered in Baltimore (Maryland, USA) with R&D offices in Kolkata, focuses on predictive maintenance and machine analytics for infrastructure systems, industrial machinery, and components including pumps, electrical supply, and compressors. They specialise in IoT solutions for pumps and

machines, electrical IoT solutions, vertical OEM solutions, and public safety solutions.

Their detection technology for vibration, temperature and electrical power is offered at a price point affordable for every industrial unit. Their platform offers 24/7 availability for online monitoring.

Vibration analysis assesses health of the rotational components of